

Intra-articular Drain Versus No Drain After Arthroscopic Anterior Cruciate Ligament Reconstruction: A Randomized, Prospective Clinical Trial

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Purpose: A significant proportion of surgeons use intra-articular drains after arthroscopic anterior cruciate ligament (ACL) reconstruction. The usual reason given to justify the use of a drain is to minimize patient pain and stiffness of the knee joint. The purpose of this study was to assess the validity of this approach. **Methods:** In this study 118 consecutive ACL reconstruction patients were randomized to no drain or a postoperative intra-articular suction drain. Inclusion criteria included a successful ACL reconstruction via either 4-strand hamstrings or bone-patellar tendon-bone autograft as a graft source. The primary outcomes were defined as pain (assessed by pain scores and analgesic counts) and range of motion (ROM) (assessed by loss of flexion and extension compared with the nonoperative leg). An independent statistical analysis was performed. **Results:** The 2 groups were comparable with respect to patient demographics, surgical findings, and procedures performed. There were no differences between the treatment groups for the primary outcomes of pain and ROM during the 8-week follow-up period. The study had adequate power to detect a clinically significant difference. Regarding the secondary outcomes, there was a difference in the grade of hemarthrosis between the groups at week 1 but not at week 4 or 8. However, the difference in subjective grade of hemarthrosis at 1 week did not have any effect on the primary outcomes of pain and ROM. During the study period, there were no complications in either group. **Conclusions:** The routine use of intra-articular drains after arthroscopic ACL surgery was not supported by this study. **Level of Evidence:** Level II, randomized controlled trial without narrow confidence intervals. **Key Words:** Anterior cruciate ligament—Suction—Drainage—Hemarthrosis—Postoperative pain—Range of motion.

Drains have been used commonly in orthopaedics to evacuate hematomas under the presumption that this will decrease pain and swelling, hasten the

return of motion, shorten the hospital stay, speed rehabilitation, and potentially decrease the risk of infection.¹⁻¹⁰ Over the years, the necessity of intra-articular drains has been questioned in the literature.¹⁻⁸ However, drains have not been abandoned for fear of local complications. Several prospective, randomized studies have evaluated the use of drains in joint arthroplasty and orthopaedic trauma, and the preponderance of these show no benefit to using drains.^{1,2,4-7} There is a paucity of literature evaluating the use of intra-articular drains after arthroscopic procedures. A prospective but nonrandomized review of 60 arthroscopic knee surgeries showed that patients with a drain and no tourniquet had fewer hemarthroses develop and had a quicker return of motion.⁹ However, this study did not include anterior cruciate ligament (ACL) patients. These patients may be different than

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other orthopaedic patients because they tend to be younger, with high functional demands. Therefore small deficits in range of motion (ROM) may have a greater effect on function and are a greater concern. In addition, with the desire to perform these as outpatient procedures, any beneficial effect from a drain in reducing postoperative pain would be of value. A study by Matava et al.¹⁰ included a questionnaire to directors of U.S. sports medicine fellowships and showed no consensus on drain use after ACL surgery, with 51% advocating routine use. Our study was preceded by an e-mail survey of Canadian sports knee surgeons, which showed a similar lack of consensus on drain use after ACL surgery. Forty-seven percent of the responding Canadian knee ligament surgeons "never" used a drain postoperatively, and thirty-three percent used a drain "routinely." The purpose of this study was to determine the efficacy of intra-articular drains in improving outcomes after arthroscopic ACL reconstruction.

METHODS

This was a prospective, randomized trial comparing the use of an intra-articular drain versus no drain after arthroscopic ACL reconstruction. The study was approved by the Clinical Investigations Committee (Fraser Health Authority, New Westminster, Canada), and informed consent was obtained from all patients. Consecutive patients with an ACL reconstruction, via either 4-strand hamstrings or a 10-mm bone-patellar tendon-bone autograft from the ipsilateral knee, were enrolled. Exclusion criteria included significant pain preoperatively (narcotic-dependent), a large swelling preoperatively (grade 3 or 4 effusion), revision cases, and patients with a known increased risk of bleeding (bleeding diathesis or prescribed anticoagulation medication).

All surgeries were performed on consecutive patients by the senior author (R.G.M.), and randomization was performed at the end of the case by opening a sealed envelope. The anesthetic type (regional or general) was not controlled and was selected at the discretion of the anesthesiologist. All patients received prophylactic intravenous antibiotics preoperatively. The procedures were all done in a standardized fashion via an arthroscopic inflow pump at 30 to 35 mm Hg. An initial diagnostic arthroscopic examination was performed with management of the menisci and articular cartilage as necessary. A tourniquet was inflated at the time of graft harvest. Graft choices consisted of 4-strand hamstrings (gracilis and semi-

tendinosus tendons doubled over) or a 10-mm bone-patellar tendon-bone autograft from the ipsilateral knee. A single-incision technique was used in all cases, and the grafts were fixed with bioabsorbable screws that were sized for the graft in both the tibial and femoral tunnels. A notchplasty was performed only if there was significant notch stenosis or if impingement of the graft was present. The amount of notchplasty was determined by the notch and graft size. In many cases the notch enlargement was minimal. Full ROM was confirmed intraoperatively, and the Lachman test was performed to confirm stability. The knee joint and portals were injected with a standard volume of bupivacaine with epinephrine (20 mL of 0.25% bupivacaine with epinephrine). In the drain group, a 1/8-inch, spring-type, disposable suction drain (Hemo-vac; Zimmer, Warsaw, IN) was inserted through the superolateral portal but not until the intra-articular bupivacaine had 5 minutes to absorb into the synovium. The drain was assumed to be intra-articular because it was inserted through the outflow cannula (which had been visualized arthroscopically), and there was immediate free backflow of bupivacaine and arthroscopy fluid from the cannula. All patients underwent application of a compression stocking and a Cryo/Cuff (Aircast, Summit, NJ) before deflation of the tourniquet. They were discharged home on the same day with a prescription for ketorolac (or acetaminophen/codeine in the presence of an allergy). Patients used the Cryo/Cuff for 1 week. All patients underwent standardized rehabilitation supervised by a physiotherapist to emphasize early ROM, immediate full extension, and weight-bearing as tolerated. Walking aides were used until there was adequate quadriceps control and a minimal limp. The drain was removed on postoperative day 1 by the home care nurse, and the amount of drainage was recorded. The patients kept a medication diary and recorded their pain on a visual analog scale (VAS) at a consistent time in the evening on postoperative days 1, 2, and 3. They were reviewed in the clinic at weeks 1, 4, and 8. Pain (VAS), ROM, grade of hemarthrosis, and thigh circumference, as well as the presence of any complications, were documented. Patients were instructed not to reveal whether they had received a drain postoperatively. Primary outcome measures included pain, measured by VAS and pill count, and ROM, measured by use of a standard handheld goniometer (extension and flexion compared with contralateral limb). Secondary outcome measures included grade of hemarthrosis, according to the classification of Coupens and Yates⁹ (graded subjectively from 0 to 4) (Table 1);

TABLE 1. *Clinical Grading of Hemarthrosis*

Grade	Description
0	No detectable fluid
1	Fluid present with fluid wave
2	Palpable fluid in suprapatellar space
3	Ballotable patella
4	Tense hemarthrosis

Hemovac drainage (in milliliters); thigh circumference (in centimeters); and the development of any complications (such as infection or wound dehiscence). An independent statistical analysis was performed by use of multiple linear regressions, Wilcoxon rank-sum tests, and multivariate analysis of variance (MANOVA), with the P value set at .05.

RESULTS

There were 121 consecutive patients but 3 declined randomization. This left 118 patients, of whom 60 were randomized to a drain and 58 were randomized to no drain. No patients were lost to follow-up.

The patient demographics were comparable for both groups (Table 2), except that the drain group had a statistically higher age (mean, 29.5 years v 25.4 years; $P = .03$, t test). There was a trend toward more male patients being included in the no-drain group, but this did not show statistical significance ($P = .07$, χ^2 test). The 2 groups were comparable with respect to important factors such as graft source, tourniquet time, and incidence of partial meniscectomy, meniscal repair, and notchplasty (Table 2).

The data collected are recorded in Table 3. For the predetermined primary outcome of pain, measured by VAS, there was no statistically significant difference between the 2 groups during the first 3 postoperative days ($P = .31$, MANOVA). In addition, there was no difference in pain at week 1 ($P = .11$, Wilcoxon rank-sum test), week 4 ($P = .83$), or week 8 ($P = .22$). Similarly, there was no difference in pill counts on days 1 to 3 ($P = .54$, MANOVA). For the second major outcome, ROM, there was no difference as measured by loss of extension at week 1 ($P = .76$, Wilcoxon rank-sum test), week 4 ($P = .87$), or week 8 ($P = .54$). In addition, there was no difference in ROM as measured by loss of flexion at week 1 ($P = .45$, Wilcoxon rank-sum test), week 4 ($P = .48$), or week 8 ($P = .62$). There was no difference in thigh circumference at week 1 ($P = .50$, multiple regression), week 4 ($P = .50$, Wilcoxon rank-sum test), or

week 8 ($P = .88$, Wilcoxon rank-sum test). The hemarthroses were statistically smaller at week 1 with use of a drain ($P = .02$, Wilcoxon rank-sum test) but not at week 4 ($P = .59$) and week 8 ($P = .86$). Although the grade of hemarthrosis was statistically lower at week 1, the clinical difference was much less than 1 grade and did not result in any change in clinical outcome with regard to pain or ROM. The mean drainage was 152 mL in the drain group. There were no infections, and 2 patients in each group had the knee aspirated before the 8-week mark.

A power calculation was performed for pain (VAS) on postoperative day 1. For the purposes of the power calculation, we considered a clinically significant difference in pain on a 10-cm VAS to be 20%. If there had indeed been a difference between the drain and no-drain groups of 2 points on our 10-point scale (20%), our study would have detected it with a probability of greater than 95%.

DISCUSSION

This study showed that for the primary outcomes of pain and ROM, the results showed no significant difference between groups. Although, as expected with a drain, there were statistically smaller hemarthroses in the drain group, this was less than 1 clinical grade and did not lead to any difference in the primary outcomes.

One of the purported advantages of using a drain is to reduce the incidence of wound problems and the risk of deep infection. However, the incidence of wound problems is small in this patient population, and although there were no wound complications in either group, this study was not designed to address

TABLE 2. *Patient Demographics*

	Drain Group	No-Drain Group	P Value
No.	60	58	
Age (y) (mean)	25.4	29.5	$P = .03$
Sex (n)			$P = .07$
Male	29	40	
Female	31	18	
Time to surgery (d) (mean)	92.8	103.0	$P = .83$
Graft source (n)			$P = .89$
Hamstrings	40	38	
Bone-tendon-bone	20	20	
Partial meniscectomy (n)	28	20	$P = .22$
Meniscal repair (n)	14	18	$P = .44$
Notchplasty (n)	37	38	$P = .96$
Tourniquet time (min) (mean)	57.8	56.4	$P = .56$

TABLE 3. Results

	Day 1		Day 2		Day 3		Week 1		Week 4		Week 8	
	Drain Group	No-Drain Group	Drain Group	No-Drain Group	Drain Group	No-Drain Group	Drain Group	No-Drain Group	Drain Group	No-Drain Group	Drain Group	No-Drain Group
Pain												
VAS	3.8	4.9	3.0	3.7	2.3	3.1	1.6	2.1	0.9	0.7	0.3	0.3
Pill counts	4.3	4.6	3.6	4.1	2.7	3.2	N/A	N/A	N/A	N/A	N/A	N/A
ROM												
Extension loss (°)	N/A	N/A	N/A	N/A	N/A	N/A	3.7	3.1	3.0	2.6	1.6	1.5
Flexion loss (°)	N/A	N/A	N/A	N/A	N/A	N/A	64.0	61.5	27.8	27.7	5.7	5.6
Thigh circumference (difference) (cm)	N/A	N/A	N/A	N/A	N/A	N/A	2.4	2.5	0.9	1.0	0.6	0.5
Hemarthrosis (grade)	N/A	N/A	N/A	N/A	N/A	N/A	2.5	2.8	1.6	1.6	0.7	0.6

N/A, Not applicable.

this issue. The rate of deep infection after ACL reconstruction is approximately 1%.^{11,12} To detect a relative risk reduction of 50% (to a 0.5% infection rate) with a power of 80%, the study would need to include more than 5,000 patients.

A post hoc analysis of the data showed that the type of ACL graft had no effect on the results. The significance of the difference in hemarthrosis between the drain and no-drain groups was $P = .59$ in the bone-patellar tendon-bone patients and $P = .23$ for the semitendinosus and gracilis grafts. A similar analysis of the data, stratifying for notchplasty, showed no difference between the drain and no-drain groups with regard to pain ($P = .39$) and hemarthrosis ($P = .46$).

There is also a question as to whether there is a difference between the patients who receive ketorolac or acetaminophen with codeine as an analgesic. This is difficult to assess in this study because 86% of the patients received ketorolac as their primary analgesia and all patients received some form of nonsteroidal anti-inflammatory for analgesia (as a preoperative suppository or intraoperative parenteral preparation). Therefore less than 10% of the overall group received acetaminophen with codeine as their primary analgesia, and they all received anti-inflammatories in the critical perioperative period.

The results of this study challenge the routine use of drains after arthroscopic ACL reconstruction. This is consistent with other studies in joint arthroplasty and orthopaedic trauma.^{1,2,4-7} Drain use seems to be guided by habit and anecdotal evidence, rather than by evidence. Avoiding drains could eliminate potential complications of drain use such as retrograde infection and painful removal. Although retrograde infection is very uncommon, patients frequently complain about the pain associated with drain removal. Not using a drain

would also obviate the need for a clinic visit on the first postoperative day and avoid the cost of the drain.

A potential limitation of this study is that neither the patients nor the examiner was blinded. Although it was not possible to blind the patient, at follow-up visits, they were instructed to not reveal their treatment group, in an effort to conceal the patients' group identity from the primary surgeon who reviewed all patients at follow-up. The strengths of this study are that it is prospective and randomized, with 1 surgeon following up a large number of consecutive patients, and that it included 2 commonly used graft sources.

We believe that routine use of intra-articular drains after uncomplicated arthroscopic ACL surgery is not supported by this study. Avoiding drains will reduce costs and eliminate the potential complications of their use.

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